Summary Report

Mobile Application Development Blackbucks Long Term Internship 2026

Name : Vanama Varshini

Roll No : 22HU1A05F3

Project Title : ChatGpt Client ChatGpt Style interface with Flutter and Riverpod

Table of Contents

Topic	Page No
Internship Overview	03 - 04
About Blackbucks	05 - 06
Course Syllabus	07 - 08
Pre-Program Phase- 01 (Week 01 – 04)	09 - 12
Foundation Phase 02 (Week 05 – 06)	13 - 14
Core Internship Phase 03 (Week 07 – 12)	15 - 19
Recap & Review Phase 04 (Week 13 – 14)	20 - 21
Project Implementation Phase (Week 15)	22
Final Phase (Week 16)	52
Conclusion	53

Internship Overview:

Total Duration: 4Months (16 Weeks)
Total Learning Hours: Up to 240 Hours
Delivery Mode: Hybrid (Live + Recorded + Platform-based Learning)
Learning Platform: Blackbucks TaPTaP LMS
Project : Individual Project selection
Communication Mode: Domain-wise WhatsApp Groups

Program Objectives

The **Blackbucks Skill Internship Program 2026** aims to:

1. Develop **industry-ready skills** through structured, domain-specific training.
2. Provide **hands-on project experience** and mentor-driven guidance.
3. Strengthen **placement and employability readiness** through mock tests and continuous evaluation.
4. Offer exposure to **modern technologies and workflows** in real-world problem-solving.
5. Enable participants to graduate with a **project portfolio and certification** validating 240 hours of learning.

Program Structure

Pre-Program Phase (Week 1 – 4)

Objective: Prepare interns for core learning and professional readiness.

Activities:

- **Daily Placement & Employability Practice Papers**
- **Pre-Assessment Test**
- **Prerequisite Courses & Recorded Content**
- **Orientation Session (2 Hours)** introducing mentors, LMS, and domains.

Foundation Phase (Week 5 – 6)

Objective: Introduce all to the selected domains through conceptual and overview sessions.

Schedule:

- **Mon / Wed / Fri:** Basics of each domain
- **Tue / Thu / Sat:** Overview and domain introductions

Weekly Activities:

- 1 Assessment (1 hr)
- 1 Assignment
- 1 Placement Test
- 1 Employability Test

Core Internship Phase (Week 7 – 12)

Objective: Strengthen technical and practical expertise.
Weekly Pattern:

- 3 Live Sessions (2 hrs each)
- 1 Assignment
- 2 Assessments
- 1 Employability Test

Recap & Review Phase (Week 13 – 14)

Objective: Revise all domain concepts and prepare for project work.

- Mentor-led recap sessions
- Domain-wise revision & short tests

Project Implementation Phase (Week 15)

Objective: Apply learning through a guided, mentor-supported project.
Details:

- 2 Mentor Support Sessions per week (6 total)
- Project divided into 5–6 steps (each step: 3 days)
- Weekly placement and employability tests continue

Final Phase (Week 16)

Objective: Assess final performance and close the internship.

- **Grand Test – Part 01 & 02 (Total 5 Hours)**
- **Project Review & Certification Ceremony**

About the Blackbucks

The Blackbucks: Pioneering Campus Success

The Blackbucks is India's premier **Campus Success Platform**, committed to transforming higher education and placement ecosystems. As a trusted partner of top **Higher Education Institutions (HEIs)**, Blackbucks provides a robust suite of technology-driven solutions designed to empower students through rigorous training, structured practice, and successful placement outcomes.

Vision: Empowering Students for a Brighter Tomorrow

At The Blackbucks, we believe in **leveraging technology to unlock the full potential** of every student. Our mission is to bridge the gap between academic learning and real-world careers by offering comprehensive support systems that align education with employability.

Key Offerings

- **TaPTaP Platform** – Advanced learning and placement tracking system
- **VPL Platform** - College domain wise labs
- **BBX Swift** – Personalized learning engine for upskilling and certification
- **University Certification Programs** – Job-linked career pathways integrated with higher education

Together, these tools provide a **360° learning and placement experience**, tailored for diverse academic backgrounds.

12 Years of Impact & Innovation

Founded in **2013**, The Blackbucks began with a revolutionary idea: *"Technology for Students."* Over the past decade, we've grown into a world-class technology company with a track record of innovation, impact, and measurable success in student outcomes.

Recognized Leaders in Post-Graduation Careers

Our **Post Graduation Career Programs**, run in collaboration with the **Government of Andhra Pradesh**, have been **awarded for seven consecutive years**, showcasing our excellence in job-linked education and guaranteed placement pathways. We take pride in offering cutting-edge training in:

- Software & Product Development
- Artificial Intelligence & Machine Learning
- Data Science & Emerging Technologies
- Cybersecurity, MBA, and Entrepreneurship Tracks

Why Choose The Blackbucks?

- **10+ years of expertise** in student-centered ed-tech innovation
- **Guaranteed placement programs** across technical and management domains
- A growing network of **200+ partner colleges and 1000+ recruiters**
- Proven success with **tens of thousands of student placements**

Join Us: Where Learning Meets Opportunity

Whether you're a **student seeking career transformation**, an **educational institution aiming to boost placement performance**, or a **recruiter looking for top talent**, The Blackbucks is your destination for success.

Company Snapshot

- **Website:** www.theblackbucks.com
- **Industry:** Technology, Information & Internet
- **Headquarters:** Hyderabad, Telangana, India
- **Company Size:** 51–200 employees
- **Founded:** 2013
- **LinkedIn Presence:** 143+ professionals currently associated

Specialties

IT & Software | Product Development | Engineering | Electronics | AI/ML | Data Science | Emerging Technologies | Higher Education | Career Services | Entrepreneurship | Jobs & Placements

Mobile Application Development – 10 Module Syllabus

Module 1: Introduction to Full Stack & Java Development

- Overview of Full Stack Development
- Web & Mobile Application Architecture
- Client–Server Model
- Development Environment Setup

Module 2: Programming Foundations

- Variables & Data Types
- Control Structures (Loops & Conditions)
- Object-Oriented Programming (OOP) Concepts
- Classes, Objects & Methods

Module 3: Frontend Development Basics

- HTML Fundamentals
- CSS Styling & Layouts
- JavaScript Basics
- Responsive Design Principles

Module 4: Data Handling & API Basics

- Introduction to APIs
- HTTP Methods & JSON
- Fetch API / AJAX
- **API Integration Basic**

Module 5: Java Web Technologies

- JavaFX / JSP Basics
- Servlets Architecture
- MVC Pattern in Java
- Java DTO Concepts

Module 6: Backend Development with Spring Boot

- Spring Boot Fundamentals
- Building REST APIs with Spring Boot
- Spring Boot Project Structure
- Dependency Injection

Module 7: Database Management

- Database Concepts
- MySQL / PostgreSQL Basics
- CRUD Operations
- Database Integration with Java

Module 8: Security & Testing

- Spring Security Basics
- Authentication & Authorization
- Unit Testing with JUnit
- API Testing

Module 9: Deployment & Cloud

- Deploy Java Apps using Tomcat
- AWS / Azure for Java Applications
- JVM & Application Performance Tuning
- Microservices & WebSockets

Module 10: Full Stack Integration & Capstone

- Frontend + Backend + Database Integration
- Building End-to-End Java Full Stack App
- Deployment & Optimization
- Capstone Project Presentation

Pre-Program Phase- 01 (Week 01 – 04) Monthly Summary Report

– December 2025

Internship Duration Covered

Month: December 2025 – Early January 2026
Program: Blackbucks Long Term Internship (LT 2026)
Focus Areas: Pre-requisite Skill Building, Company-Based Evaluation, Employability Assessments

---

Overview of the Internship Activities

The Blackbucks Long Term Internship (LT 2026) is designed to prepare students for industry readiness through a structured approach involving **technical foundation strengthening**, **company-aligned mock assessments**, and **employability evaluations**.
During December 2025, the program focused on completing pending pre-requisite topics and periodically evaluating students' progress through mock tests and employability assessments.

Module 1: Pre-Requisites Learning (Skill Foundation Module)

Duration

- Conducted across **Weeks 01 to 04**
- Multiple days each week based on pending course requirements

Description

This module focused on strengthening students' **core technical and conceptual foundations** required for successful participation in the internship and future industry roles. Students continued learning pending syllabus topics aligned with their selected domain.

Key Learning Outcomes

- Completion of balance technical topics from the assigned course
- Improved conceptual clarity and problem-solving skills

- Readiness for company-level assessments and real-world tasks
- Reinforcement of fundamentals required for advanced internship activities

Mode

- Self-paced and guided learning
- Progress monitored through scheduled assessments

Module 2: Company-Based Mock Tests

Schedule

- **Mock Test 01:** 9th December 2025
- **Mock Test 02:** 16th December 2025
- **Mock Test 03:** 23rd December 2025
- **Mock Test 04:** 30th December 2025
- **Timing:** 6:00 PM – 11:00 PM

Description

Company-based mock tests were conducted to simulate **real corporate hiring assessments**. These tests were designed based on actual company recruitment patterns to familiarize students with industry expectations.

Key Objectives

- Evaluate technical, aptitude, and logical reasoning skills
- Prepare students for real placement and internship tests
- Improve time management and assessment handling skills
- Identify skill gaps and areas for improvement

Outcome

Students gained hands-on experience with company-style evaluations, enhancing their confidence and assessment readiness.

Module 3: Employability Assessments

Schedule

- **Assessment 01:** 13th December 2025
- **Assessment 02:** 20th December 2025
- **Assessment 03:** 27th December 2025
- **Assessment 04:** 3rd January 2026
- **Timing:** 6:00 PM – 11:00 PM

Description

Employability assessments were conducted to measure students' overall **job-readiness**, beyond technical skills. These assessments focused on aptitude, analytical thinking, communication ability, and professional awareness.

Key Evaluation Areas

- Problem-solving and analytical ability
- Logical reasoning and aptitude skills
- Industry readiness and employability parameters
- Performance consistency across assessments

Outcome

The assessments provided a structured benchmark of student readiness and helped track progressive improvement throughout the internship period.

Overall Internship Outcomes (December 2025)

- Successful continuation and completion of pending pre-requisite learning modules
- Exposure to **four company-based mock tests**
- Participation in **four employability assessments**
- Enhanced technical competency and assessment confidence
- Improved industry alignment and placement preparedness

Conclusion

The December 2025 phase of the Blackbucks Long Term Internship (LT 2026) provided a structured and outcome-oriented learning experience. Through continuous learning, regular mock testing, and employability assessments, students demonstrated progressive improvement in both technical and professional competencies.

This month served as a strong foundation for advanced internship activities and real-world industry engagement in subsequent phases.

Foundation Phase 02 (Week 05 – 06)

Blackbucks Long Term Internship (LT 2026)
Live Sessions Summary Report – Mobile Application Development
Period Covered: January 2026

Phase Overview

The Fundamental Phase focused on building core programming foundations, web development concepts, and initial placement readiness. This phase emphasized Full Stack development fundamentals, JavaScript programming, Mobile basics, and early exposure to company-based assessments.

Week 05: Introduction to Full Stack & Programming Foundations

Duration: 5th January 2026 – 10th January 2026

Key Activities

- Internship onboarding and orientation
- Introduction to Mobile session
- Programming Foundations session (OOP, Variables, Loops)
- Session-wise assessments
- TCS NQT Placement Mock Test
- Employability Test – 01

Learning Focus

- Overview of Mobile application development
- Understanding programming fundamentals
- Object-Oriented Programming concepts
- Variables, loops, and basic program logic
- Familiarization with placement-style assessments

Week 05 Outcomes

- Clear understanding of full stack development fundamentals
- Improved logical and programming thinking
- Basic knowledge of OOP concepts and program structure
- Initial benchmarking of technical and aptitude skills

Week 06: Core Programming Concepts & Application Logic

Duration: 12th January 2026 – 17th January 2026

Key Activities

- Session on core programming concepts
- Application logic and structured coding session
- Lesson plan-based learning and revisions
- Session-wise assessments

Learning Focus

- Control structures and program flow
- Writing structured and modular programs
- Applying programming concepts to real-world applications
- Strengthening coding logic and debugging skills

Week 06 Outcomes

- Conceptual clarity in core programming concepts
- Ability to build structured program logic
- Improved problem-solving and coding efficiency

Phase 03: Core Concept Phase (Week 07 – 12)

Duration: Week 03 onwards (February 2026)

Phase Overview

The Core Concept Phase focused on advanced mobile application development concepts, backend integration, database management, and deployment practices. This phase strengthened practical mobile development skills and enhanced placement readiness through continuous assessments and hands-on learning.

Week 07: Data Handling, APIs & Version Control

Duration: 19th January 2026 – 24th January 2026

Key Activities

- Session on Data Handling & APIs Basics
- Assessment 02
- Placement Mock Test (3 Hours Duration)
- Session on Version Control & Project Setup (Git, GitHub)
- Assessment 03
- Placement Mock Test (3 Hours Duration)

Learning Focus

- Data handling techniques in web applications
- API concepts and request-response architecture
- Version control using Git and GitHub
- Project setup and collaborative development workflows

Week 07 Outcomes

- Ability to work with APIs and structured data
- Understanding of version control workflows
- Improved project management and collaboration skills
- Enhanced readiness for placement-based assessments

Week 08: Mobile UI Design & Backend Integration

Duration: 2nd February 2026 – 7th February 2026

Key Activities

- Placement Mock Test (3 Hours Duration)
- Assignment 2
- Session on Mobile UI Components & Layouts
- Assessment
- Employability Test
- Session on Backend for Mobile Apps (Firebase / REST APIs)

Learning Focus

- Designing mobile user interfaces using components and layouts
- Mobile UI/UX principles and responsive design
- Backend integration using Firebase and REST APIs
- Handling data communication between mobile apps and servers

Week 08 Outcomes

- Ability to design responsive mobile application interfaces
- Understanding backend connectivity for mobile apps
- Improved UI development and integration skills
- Strengthened placement readiness through assessments

Week 09: Mobile Databases & Authentication

Duration: 9th February 2026 – 14th February 2026

Key Activities

- Placement Mock Test (3 Hours Duration)
- Employability Test
- Session on SQLite / Cloud Databases
- Assessment
- Assignment 3
- Session on Mobile App Authentication

Learning Focus

- Local database management using SQLite

- Cloud database integration for mobile applications
- Authentication and authorization in mobile apps
- Secure user login and data handling

Week 09 Outcomes

- Ability to manage mobile application data storage
- Understanding of authentication mechanisms
- Improved knowledge of secure mobile app development
- Enhanced backend integration capabilities

Week 10: State Management & API Integration

Duration: 16th February 2026 – 21st February 2026

Key Activities

- Assessment
- Placement Mock Test (3 Hours Duration)
- Session on State Management & Navigation
- Employability Test
- Assignment 4
- Session on API Integration in Mobile Apps

Learning Focus

- Application state management techniques
- Navigation and screen transitions in mobile apps
- Integrating external APIs for dynamic data
- Managing application data flow and UI updates

Week 10 Outcomes

- Ability to manage app states and navigation flows
- Practical experience with API integration
- Improved mobile app architecture understanding
- Strengthened development and debugging skills

Week 11: Testing & Deployment of Mobile Applications

Duration: 23rd February 2026 – 28th February 2026

Key Activities

- Assessment
- Placement Mock Test (3 Hours Duration)
- Session on Mobile App Testing & Debugging
- Employability Test
- Assignment 5
- Session on Mobile App Deployment (Play Store / App Store)

Learning Focus

- Debugging techniques for mobile applications
- Mobile application testing strategies
- Preparing applications for deployment
- Publishing apps to Play Store and App Store

Week 11 Outcomes

- Ability to test and debug mobile applications
- Understanding of mobile app deployment processes
- Improved application stability and performance
- Enhanced industry readiness

Week 12: Advanced Mobile Features & Performance Optimization

Duration: 2nd March 2026 – 7th March 2026

Key Activities

- Assessment
- Placement Mock Test (3 Hours Duration)
- Session on Cloud Services, Push Notifications & Sensors
- Employability Test
- Assignment 6

- Session on Mobile App Performance & Mobile UI/UX + Backend

Learning Focus

- Integrating cloud services into mobile apps
- Implementing push notifications and device sensors
- Mobile application performance optimization
- Advanced UI/UX improvements and backend integration

Week 12 Outcomes

- Ability to integrate advanced mobile features
- Improved mobile application performance optimization skills
- Enhanced understanding of full mobile application lifecycle
- Strengthened mobile development skills and industry readiness

---

Phase 04: Recap & Review Phase (Week 13 – 14)

Duration: Final Weeks (March 2026 onwards)

Final Project :

ChatGpt Client Chatgpt Style interface with Flutter and Riverpod

Abstract:

A ChatGpt –Style interface built with Flutter and Riverpod is a modern approach to creating interactive and intelligent chat applications. Flutter is a Powerful UI framework developed by Goggle that allows developers to build fast and attractive mobile application using a single codebase. Riverpod is a state management tool is used in Flutter that helps manage application data effectively and keeps the code organized and scalable. By combining Flutter's flexible user interface with Riverpod's effectively state management developers can create a smooth chat interface similar to ChatGpt .

The main goal of the project is to design a clean, responsive and user-friendly chat application that demonstrates how modern technologies can be used in many applications such as customer support boats ,AI assistants and learning platforms.

Introduction:

Many mobile applications need a simple and interactive chat system for communication between users and AI assistants. However, building a chat interface that is fast, organized and easy to manage can be difficult.

- Chat applications are used for communication
- Earlier chat apps only sent and received messages
- Traditional chat systems did not use AI
- Old systems had simple UI and manual state handling
- Modern chat apps use AI for automatic responses
- Flutter is used to build the mobile app UI
- Riverpod is used to manage app state
- APIs are used to send and receive chat data

- Modern apps are faster and more interactive

Problem Statement:

Problems such as slow updates, poor message handling and complex code structure often occur in traditional applications. Therefore ,there is a need to develop a Chatgpt-Style chat interface that provides smooth messaging and efficient data management using modern technologies like Flutter and Riverpod.

Objectives:

- To develop a chatgpt-style mobile application usingFlutter
- To implement a responsive and interactive chat user interface.
- To use Riverpod for efficient state management in the application.
- To design a clean,user-friendly and modern UI/UX.
- To integrate AI API for generating intelligent responses.
- To manage chat data dynamically with in the app.
- To enable real-time communication between user and AI system.

Scope of the Project:

In Scope

- User login and chat interface development
- Sending user messages and receiving AI responses
- Managing chat data using Flutter and Riverpod

Out of Scope

- Voice chat and video call features
- Multi-language translation support
- Advanced cloud security and payment integration

Literature Review:

Many chat applications are developed using mobile frameworks like Flutter because it supports Android, iOS, web, and desktop with a single codebase.Riverpod is widely used for state managemen because it is simple,scalable, and updates the UI efficiently.Traditional chat applications only support sending and receiving messages between users.Modern AI-based chat systems use APIs to generate intelligent responses automatically. OpenAI and similar AI services are commonly used to provide chatbot functionality.

System Architecture:

- User enters message in chat screen
- Flutter UI sends the message
- Riverpod manages the chat state
- API service sends request to backend
- Backend processes the message
- Database stores chat details if needed
- AI response is returned to the app
- Flutter UI displays the response to the user

Architecture Flow:

User

↓

Mobile App (Flutter UI)

↓

Riverpod State Management

↓

API Service

↓

Backend Server

↓

Database / AI Model

↓

Response Returned

↓

Display Response to User

Tools And Technologies:

Flutter

- Used to develop the mobile application UI

Dart

- Programming language used in Flutter

Riverpod

- Used for state management

VS Code / Android Studio

- Used to write and run the code

HTTP Package

- Used to send API requests and receive responses

OpenAI API

- Used to generate AI responses

GitHub

- Used to store and manage source code

Android Emulator / Mobile Device

- Used to test the application

Firebase / Database

- Used to store user and chat data if needed

System Design:

- Login Screen
- Home Screen
- Chat Screen
- Riverpod State Mnagement
- API Service
- Database
- Navigation Flow

Login Screen → Home Screen → Chat Screen → Logout

Implementation:

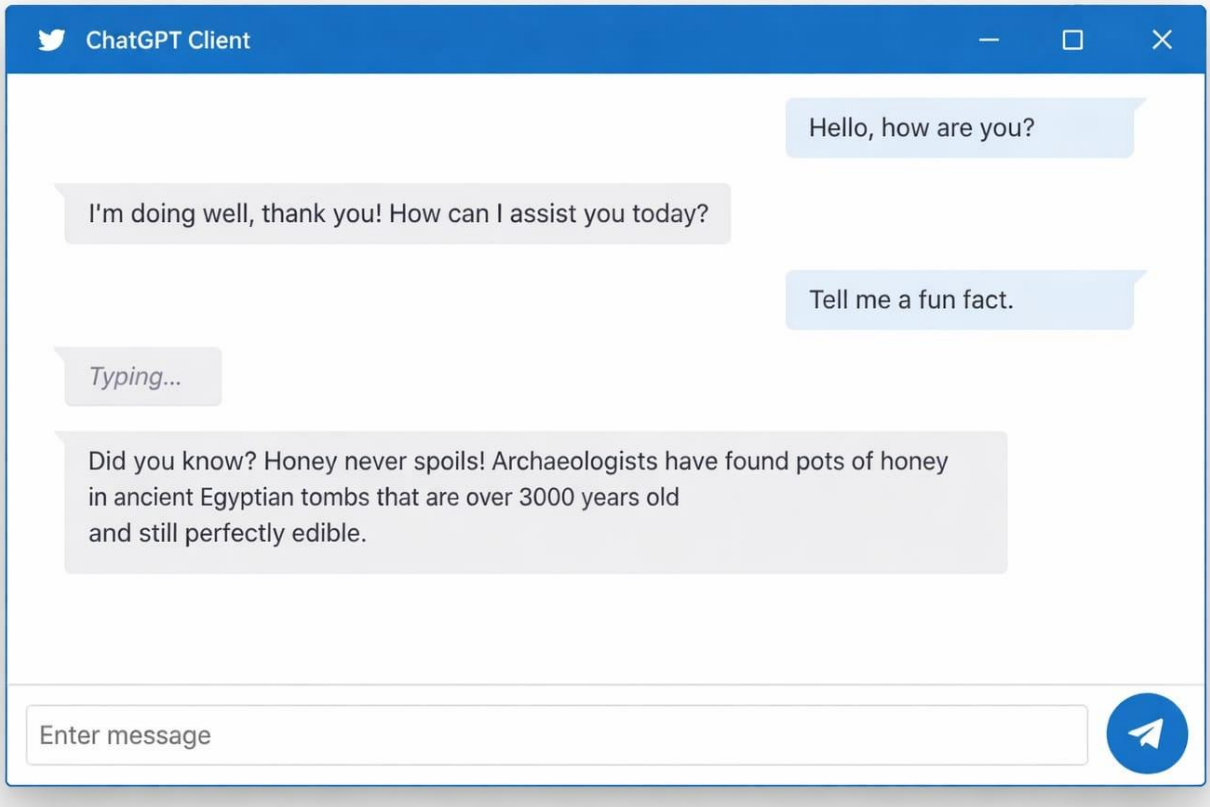
- Flutter is used to create the mobile application interface
- Dart is used as the programming language
- Riverpod is used for state management
- User login screen is created with email and password fields
- Chat screen is developed using TextField, ListView, and Send Button
- Messages are stored using a Message model class

- API service is used to send user messages to the backend
- HTTP package is used for API communication
- AI responses are displayed automatically on the screen
- Error messages are shown if login or API request fails
- Navigation is implemented between login screen and chat screen
- Chat data can be stored in Firebase or local database if needed

Data Processing WorkFlow:

- User enters message
- System checks the message
- Message is validated
- Valid message is sent to API
- API processes the message
- AI generates response
- Response is received by app
- Response is stored in chat list
- Response is displayed to user

Output Results:



Observations:

- Riverpod updates the UI automatically when messages are added
- API communication is fast and gives real-time responses
- Flutter makes the app work smoothly on mobile and laptop
- The chat history can be managed easily
- Error handling helps when API or login fails

Prototype Development:

- Created a basic login screen and chat screen
- Developed message sending and receiving functionality

- Used Riverpod for state management
- Connected the app with API service
- Tested the prototype on mobile and laptop

Testing:

- Tested login functionality
- Tested message sending and response receiving
- Checked API error handling
- Verified UI updates after every message
- Tested application on different screen size

Performance Analysis:

- Application gives fast response for messages
- Riverpod updates UI efficiently
- Flutter provides smooth user experience
- Low memory usage during chat process
- Good performance on both mobile and laptop

Scalability:

- New features can be added easily
- Database support can be extended
- Multiple users can use the app
- Cloud services can improve scalability
- More APIs can be integrated in fut

Reliability:

- Chat messages are displayed correctly
- App continues to work even after multiple messages
- Error handling improves reliability
- Login validation prevents wrong input
- Stable performance during testing

Security Considerations:

- User login protects application access
- API keys should be stored securely
- Passwords should be encrypted
- User data should be protected
- Database access should be restricted

Challenges Faced:

- Managing API responses correctly
- Handling login and validation errors
- Updating UI dynamically using Riverpod
- Maintaining smooth performance
- Debugging Flutter package issues

Future Enhancement:

- Add voice input support
- Add dark mode feature

- Add multi-language support
- Store chat history in cloud database
- Add user profile and settings page

Applications:

- AI chatbot applications
- Customer support systems
- Educational chat applications
- personal assistant apps
- Business communication tools

Limitations:

- Requires internet connection
- Depends on API availability
- Limited offline support
- API usage may have cost
- Basic version has limited features

Advantages:

- Simple and easy to use
- Fast response time
- Attractive user interface
- Easy to maintain and update
- Works on mobile, web, and desktop

Deployment Architecture

Mobile Device / Laptop → Internet → Cloud Servers → Database → API Service → Response

- Mobile Device / Laptop Used by the user to access the application
- Internet Connection
- Connects the app to the server
- Cloud Server
- Processes user requests and API calls
- Database Server
- Stores user details and chat history
- API Service
- Sends and receives data between app and backend
- *AI Service*
- Generates responses for user messages
- Authentication Service
- Validates user login details

Realistic Architecture Diagram :

1. DATASET STRUCTURE DIAGRAM :

```
+---------------------+
|    ChatMessage      |
+---------------------+
| id : String (PK)    |
| message : String    |
| role : String       |
```

```
| timestamp : DateTime|
| isLoading : Boolean |
+--------------------+
| belongs to        |
+--------------------+
|    Conversation     |
+--------------------+
| conversationId (PK) |
| title               |
| createdAt           |
+--------------------+
```

2. CLASS DIAGRAM (Flutter + Riverpod) :

```
+----------------------+
|   ChatMessage        |
+----------------------+
| id : String          |
| message : String     |
| role : String        |
| timestamp : DateTime |
+----------------------+
+----------------------+
|   ChatNotifier       |
+----------------------+
| List<ChatMessage>    |
+----------------------+
| addMessage()         |
| removeMessage()      |
| setLoading()         |
+----------------------+
+------------------------------+
| StateNotifierProvider        |
+------------------------------+
| Provides ChatNotifier State  |
```

3. DATA FLOW DIAGRAM (DFD – LEVEL 1) :

```
+----------------+

|     User       |

+----------------+
```

```
        |
        v
+----------------+
|  Flutter UI    |
+----------------+
        |
        v
+--------------------+
| Riverpod State     |
+--------------------+
        |
        v
+----------------+
| Response       |
+----------------+       ----------------------
|                  |
v                  v


+--------------+    +----------------+
| Local Storage |    | API Service     |
+--------------+    +----------------+
                        |
```

```
                    v
              +----------------+
              | AI Server      |
              +----------------+
                    |



                    |
                    v
        +--------------------+
        | Riverpod State Update|
        +--------------------+
              |
              v
        +----------------+
        | Display in UI  |
              +----------------+
```

4. SEQUENCE DIAGRAM :

```
User     UI     Riverpod    API Service    Server
  |      |        |           |            |
  |--Message->|        |           |            |
```

```
|        |--update-->|          |          |
|        |        |--request--> |          |
|        |        |          |--process--> |
|        |        |          |<--response-- |
|        |        |<--data------ |          |
|        |<--update---|          |          |
|<--display-|        |          |          |
```

5. SIMPLE VISUAL DIAGRAM (BEST FOr REAL)

```
[User]
↓
[Flutter UI]
↓
[Riverpod State]
↓
[API Call]
↓
[AI Server]
↓
[Response]
↓
```

[Update State]

↓

[Display chat]

REAL-WORLD USE CASES

1. Education & Learning

Students can use the app as a **personal tutor**.

Use Cases:

- Ask subject-related doubts (Math, Science, Programming)
- Get explanations in simple language.
- Generate notes and summaries

Example:

A student asks: "Explain Flutter widgets" → App provides structured explanation.

2. Software Development Assistance

Developers can use it for coding help and debugging.

Use Cases:

- Generate Flutter UI code
- Fix errors in Dart code

- Learn Riverpod concepts

Example:

User inputs: "Fix this Flutter error" → App suggests corrected code.

3. Content Creation

Helps writers and creators generate content quickly.

Use Cases:

- Blog writing
- Social media captions
- Report generation

Example:

User asks: "Write a project abstract" → App generates content instantly.

4. Customer Support Automation

Businesses can integrate the app for **automated chat support**.

Use Cases:

- Answer FAQs
- Provide product info

- Handle customer queries 24/7

Example:

User asks: "What are your store timings?" → Instant response.

5. Personal Productivity Assistant

Acts as a **daily assistant** for users.

Use Cases:

- To-do list suggestions
- Time management tips
- Quick information lookup

Example:

User: "Plan my day" → App generates schedule.

6. Language Translation & Learning

Useful for multilingual communication.

Use Cases:

- Translate text
- Learn new languages
- Grammar correction

Example:

User: "Translate to Hindi" → App gives accurate translation.

7. Healthcare Information (Basic)

Provides general health-related guidance (non-diagnostic).

Use Cases:

- Symptoms explanation
- Health tips
- Diet suggestions

Example:

user: “Tips for healthy diet” → App responds with guidelines.

8. Business & Office Use

Improves workplace productivity.

Use Cases:

- Draft emails
- Generate reports
- Brainstorm ideas

Example:

User: “Write a formal email” → App generates email draft**.**

9. Interview Preparation

*Helps job seekers prepare effectively**.***

Use Cases:

- Mock interview questions
- Resume suggestions
- HR answers

Example:

User: “Common HR questions” → App lists answers**.**

10. Entertainment & Casual Use

Used for fun and engagement.

Use Cases:

- Jokes and stories
- Casual conversations
- Games (quiz, riddles)

Example:

User: “Tell me a joke” → App responds instantly**.**

Advantages in Real World

- 24/7 availability
- Fast responses
- Scalable solution
- User-friendly interface
- Cross-platform (Android & iOS)

Problem VS Solution Mapping:

No chat system	**Added chat screen**
Difficult state handling	Used Riverpod

No AI response	Added API service
Login errors	Added validation
No chat storage	Added database support
Slow response	Used fast API calls
Hard to manage code	Used folder structure
No security	Added login system

Data Model/Scheme :

Name Table	Table
User	user_id, username, email, password
chat	chat_id, user_id, created_date
Message	message_id, chat_id, sender, text, timestamp
Login	login_id, email, password

Risk Analysis:

Risk	Solution
API failure	Show error message
Wrong login details	Add validation
Internet issue	Show network error
Data loss	Store data in database

Slow response	Optimize API calls

Glossary:

Term	Meaning
Flutter	Framework used to build mobile apps
Dart	Programming language used in Flutter
Riverpod	State management tool
API	Used to connect app with backend service
Backend	Server-side part of the application

System Requirements:

Hardware Requirements

- Laptop or Desktop
- Android Mobile Device
- Minimum 4 GB RAM
- Minimum 20 GB Storage
- Internet Connection

Software Requirements

- Flutter SDK
- Dart SDK
- VS Code or Android Studio
- Riverpod Package

- HTTP Package
- OpenAI API Key
- Android Emulator or Physical Device
- Windows Operating System / Android OS

Methodolgy:

Requirement Analysis

- Identify project goals and features
- Understand user needs

System Design

- Create architecture diagrams
- Design UI screens and database structure

Development

- Build login screen and chat screen
- Use Flutter and Riverpod for implementation

API Integration

- Connect app with AI API
- Send and receive chat messages

Testing

- Test login, chat, and API response
- Fix errors and improve performance

Deployment

- Run app on mobile or laptop
- Make application ready for users

Data Quality Management:

- Ensures data is correct and accurate
- Helps avoid wrong responses and errors
- Improves user experience
- Makes the application more reliable
- Keeps chat history organized
- Helps in better decision making
- Reduces duplicate or missing data

Data Dimensions:

- Accuracy
  - Data should be correct and error-free
- Completeness
  - All required fields should be filled
- Consistency
  - Data should be the same in all places
- Timeliness
  - Data should be updated and available on time
- Uniqueness
  - Duplicate data should be avoided

- Validity
    - Data should follow the required format

Data Validation

- Check if email field is empty
- Check if password field is empty
- Validate correct email format
- Ensure password has minimum length
- Prevent duplicate user entries
- Validate message input before sending
- Check API response before displaying output

Monitoring and Logging

- Monitoring is used to track application performance
- Logging is used to record system activities and errors
- Helps identify API failures and login issues
- Useful for debugging and maintenance
- Improves system reliability and performance

Monitoring Tools

- Firebase Crashlytics
- Google Analytics
- Flutter DevTools

- Logcat
- Cloud Monitoring Tools

Alerts and Notifications

- Send alert when login fails multiple times
- Notify user when internet connection is lost
- Show message when API request fails
- Alert admin if server is down
- Notify when database storage is full

Backup And Diasaster Recovery:

- Backup is used to save important user and chat data
- Data can be stored in cloud storage or database backup
- Regular backup helps prevent data loss
- Disaster recovery helps restore the system after failure
- Backup should be taken daily or weekly
- User login data and chat history should be protected
- Cloud backup improves reliability
- Recovery plan helps restart the app quickly after crash
- Database backup can restore deleted or lost records
- Backup and recovery improve system security and availability

Ethical Considerations:

- User data should be kept private

- Passwords should be stored securely
- API keys should not be shared publicly
- User chat history should be protected
- The application should avoid generating harmful content

Comparative Analysis:

- Traditional chat apps only support normal messaging
- Modern AI chat apps provide automatic responses
- Flutter provides better UI compared to older technologies
- Riverpod is easier and more efficient than manual state management
- AI-based chat systems are faster and more interactive

Learning Outcomes:

- Learned Flutter app development
- Learned Riverpod state management
- Learned API integration
- Learned UI design and navigation
- Learned testing and debugging techniques

Phase Overview

The Recap & Review Phase was designed to consolidate all the knowledge and skills acquired throughout the program. This phase emphasized revision of key concepts, strengthening weak areas, and ensuring readiness for real-world challenges and placement opportunities.

Week 13: Mobile App Integration & Performance Concepts

Duration: 9th March 2026 – 14th March 2026

Key Activities

- Placement Mock Test *(3 Hours Duration)*
- Session on Cloud Services, Push Notifications & Sensors
- Employability Test
- Practice
- Session on Mobile App Performance, Mobile UI/UX & Backend Integration

Learning Focus

- Understanding cloud service integration in mobile applications
- Learning the role of push notifications and mobile device sensors
- Exploring mobile app performance optimization techniques
- Introduction to mobile UI/UX concepts and backend connectivity
- Continuous assessment and placement readiness preparation

Week 13 Outcomes

- Ability to understand and apply mobile app integration concepts
- Improved knowledge of cloud services, notifications, and sensors in app development
- Better understanding of mobile app performance and UI/UX principles
- Strengthened technical confidence through assessments and employability preparation
- Enhanced readiness for real-world mobile application development workflows

Week 14: Project Development & Evaluation Duration:

16th March 2026 – 21st March 2026

Key Activities

- Session on Project Phase
- Employability Test
- Placement Mock Test *(3 Hours Duration)*
- Holiday
- Assignment 5
- Session on Project Phase

Learning Focus

- Continued implementation of mobile application development concepts through project-based learning

- Strengthening understanding of app development workflow and project execution
- Enhancing problem-solving and practical implementation skills
- Continuous placement preparation through mock tests and employability assessments
- Reinforcing learning through assignments and structured project sessions

Week 14 Outcomes

- Ability to apply mobile application development concepts in project tasks
- Improved understanding of app development workflow and execution
- Strengthened technical, analytical, and implementation skills
- Better confidence in handling project-based mobile application scenarios
- Enhanced readiness for placement and real-world development environments

---

Phase 05: Project Implementation Phase (Week 15)

Duration: Mid March 2026 onwards

Phase Overview

The Project Implementation Phase focused on applying mobile development skills in real-world scenarios, emphasizing app design, development, and deployment.

Week 15: Mobile App Finalization & Presentation

Duration: 23rd March 2026 – 28th March 2026

Key Activities

- Session on Project Phase
- Employability Test
- Placement Mock Test (3 Hours Duration)
- Assignment 5
- Session on Project Phase

Learning Focus

- Finalization of mobile applications
- Performance tuning and bug fixing
- Integration of all app features
- Testing, validation, and refinement
- Preparation for technical interviews and assessments

Week 15 Outcomes

- Ability to finalize and present mobile applications
- Improved app optimization and debugging skills
- Strong understanding of end-to-end app development
- Enhanced confidence and readiness for job roles

Phase 06: Final Phase (Week 16)

Duration: Week 16 (March 2026 – Ending Phase)

Phase Overview

The Final Phase focused on completing all documentation and submissions, ensuring a professional conclusion to the internship.

Key Focus Areas

- Final project report preparation and submission
- Documentation of all phases and learnings
- Portfolio and resume finalization
- Submission of internship deliverables
- Feedback and evaluation

Outcome

Students successfully completed and submitted their work, showcasing a well-structured portfolio aligned with Mobile Application Development industry standards.

Overall Phase-Wise & Week-Wise Outcomes

- Structured progression from Java web development to advanced backend architecture
- Continuous assessments and skill benchmarking
- Hands-on exposure to Spring Boot, MySQL, REST APIs, and cloud deployment
- Practical experience in authentication, testing, and microservices
- Strong placement-oriented preparation and improved professional readiness

---

Lesson Plan Link :
https://taptap.blackbucks.me/lessonPlan/?lessonPlanId=73244&testType=collegeL essonPlan

Conclusion

The phase-wise structure of the **Blackbucks Long Term Internship 2026 – Mobile Application Development** ensured systematic skill development. From the **Fundamental Phase (Phase 02)** to the **Core Concept Phase (Phase 03)**, students built strong analytical foundations and practical expertise.

The **Recap & Review Phase (Phase 04)** strengthened understanding through revision and interview preparation, while the **Project Implementation Phase (Phase 05)** provided hands-on experience with real-world projects, ensuring complete industry readiness